

Aayush Pandey

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EDUCATION

Arizona State University

Bachelor of Science in Electrical Engineering

Graduation: May 2028

EXPERIENCES

Kinective — Software Development Engineering Intern

June 2026 - Present

- Created scalable Java and Spring Boot integrations linking Kinective's platform to Corelation KeyBridge, a credit union banking system
- Authored automated Cucumber and Kotlin test suites spanning 20+ core banking workflows, driving release coverage past 80% reliably
- Resolved GraphQL error handling and phone normalization defects, preventing production failures across core-banking integrations
- Automated GitLab CI/CD quality gates for Kotlin and Cucumber suites, making backend releases faster and safer across environments

Future Today Inc. — Software Engineering Intern

Aug 2025 - Dec 2025

- Maintained real-time content sync for 5M+ HappyKids users leveraging Redis caching, AWS S3/CDN, and GitHub Actions CI/CD
- Built a Python/Pytest validation pipeline for catalog edits, blocking 100+ defects before release across connected streaming platforms
- Corrected 10,000+ MySQL records and enforced metadata rules before QA, omitting rework and reducing validation cycle time by 30%
- Wrote Python validators flagging missing fields and release-blocking errors across Roku, Fire TV, mobile, and connected TV catalogs

Arizona State University — Software Engineer Researcher

Jan 2024 - May 2024

- Parallelized data loading and preprocessing across 8+ A100 GPUs, increasing model-training throughput by 84.3% for concurrent runs.
- Analyzed 6,000 experiments and found $p=1.8$ led DoWG settings at 71.77% CIFAR-10 accuracy, informing future optimizer tuning
- Built deterministic result checks across 30+ runs, sustaining 99% output consistency and improving cross-run comparison accuracy by 50%

PROJECTS

Gene Editing Clinical Trial Intelligence Platform | [GeneEditing Website](#)

Feb 2026 - May 2026

- Architected a Next.js API route that normalized 303+ ClinicalTrials.gov records into 50-record pages, surfacing 18 actively recruiting trials, creating a full-stack dashboard adopted by 100+ researchers
- Served API-backed filters for status, phase, modality, sponsor, target, and location, simplifying clinical-trial discovery workflows
- Powered a fully serverless backend with AWS Lambda, API Gateway, DynamoDB, and S3, scheduling EventBridge jobs for automatic refresh
- Implemented CloudWatch monitoring and EventBridge-driven tracking of eligibility criteria, recruitment status, and sponsor changes
- Built trial search, comparison, and CSV export tools, streamlining cross-study analysis for CRISPR and gene therapy research efforts

SlackDB — 1st Place Winner at VillageHacks | [GitHub](#)

April 2026

- Built SlackDB, a Slack-native approval tool letting engineers review and approve risky SQL migrations before production deployment
- Load-tested 6,000 API requests at 20 requests/second with p95 latency below 500 ms and under 1% errors using FastAPI and Slack Bolt
- Integrated AutoDB AI APIs to score SQL risk, generate rollback plans, and require a second approver for high-risk migrations in Slack
- Aggregated 8 migration KPIs through FastAPI/audit endpoint, cutting manual approval reporting time by 40% across engineering teams
- Containerized the complete deployment with Docker Compose, PostgreSQL, and Redis, streaming live risk metrics to a Vercel dashboard

Open Source Contributor — Microsoft Kiota | [GitHub](#)

June 2026

- Merged a C#/ .NET fix into Microsoft Kiota that logs failing OpenAPI paths while preserving parser exception behavior in production
- Added xUnit regression tests for x-ms-enum-flags parsing, then validated edge-case handling with targeted Kiota builder coverage
- Iterated on maintainer feedback to align parser error handling, test naming, and regression coverage with Kiota contribution standards

NES 6502 Emulator | [GitHub](#)

Jan 2026 - April 2026

- Engineered a fully cycle-accurate NES emulator in Rust that boots real game cartridges, replicating authentic 1980s hardware behavior
- Implemented every 6502 CPU instruction, addressing mode, and interrupt type, wiring DMA stalls and all memory-mapped I/O registers
- Rendered 256x240 frames via the pixels and winit Rust crates, validating every CPU instruction trace against nestest execution logs
- Parsed iNES cartridge headers and implemented Mapper 0 support, accurately decoding PRG/CHR ROM banking and cartridge memory layout
- Modeled CPU status flags, stack operations, and per-instruction cycle timing to guarantee deterministic, hardware-accurate execution

Modified DoWG Optimizer Characterization | [Notebook](#)

Aug 2025 - May 2026

- Built a modified DoWG optimizer study to compare CIFAR-10 training behavior across controlled ResNet18 experiments and sweeps
- Implemented the optimizer as a custom PyTorch Optimizer with EMA weight swaps, adaptive step sizing, and CUDA tensor updates
- Ran deterministic ResNet18 experiments with torchvision transforms, DataLoader batching, seeded runs, and reusable logging
- Plotted loss and accuracy sweeps with NumPy, Matplotlib, and tqdm logs to compare convergence curves against baseline optimizers

SKILLS & INTERESTS

Languages/Tools: Python (Pandas, NumPy), Java, Kotlin, JavaScript, TypeScript, Rust, C, C++, SQL

Technical: React, FastAPI, Spring, PyTorch, AWS (Lambda, S3, DynamoDB, API Gateway, CloudWatch, EventBridge), Docker, Git, MySQL, Redis, Vercel

Interests: Anime, Weightlifting, Running, Skateboarding, Gaming (Valorant, Fortnite), NBA, Cooking